

ISYE 6740 Summer 2026
Homework 3
(100 points + 12 bonus points)

Student Name Here

This assignment does not have any Gradescope requirements.

Provided Data:

- Q2: Implementing EM for MNIST Dataset [data.dat, data.mat, label.mat, label.dat]

1. Concept Questions (10)

1. (5 points) For the EM algorithm for GMM, show how to use the Bayes rule to derive τ_k^i in a closed-form expression.
2. (5 points) Based on the outline given in the lecture, derive the maximum likelihood estimators (MLEs) for the mean and covariance of an n -dimensional Gaussian random variable. Assume m *i.i.d.* (independent and identically distributed) observations $x^1, \dots, x^m$ that follow a $\mathcal{N}(\mu, \Sigma)$ distribution. Show that maximum likelihood estimates of the mean and variance parameters are

$$\hat{\mu} = \frac{1}{m} \sum_{i=1}^m x^i, \quad \hat{\Sigma} = \frac{1}{m} \sum_{i=1}^m (x^i - \hat{\mu})(x^i - \hat{\mu})^T,$$

respectively. Please show the work for your derivations in full detail. Discuss: what is the rank of $\hat{\Sigma}$ when $m < n$? In the extreme case, when $m = 1$ (i.e., only one sample), what is the rank of $\hat{\Sigma}$? This question shows that, when the data is high-dimensional, the sample size needs to be sufficiently large relative to the dimension (otherwise we need special treatment to handle low-rank covariance matrix).

2. Implementing EM for MNIST dataset (25 points)

Implement the EM algorithm for fitting a Gaussian mixture model for the MNIST handwritten digits dataset. For this question, we reduce the dataset to be only two cases, of digits “2” and “6” only. Thus, you will fit GMM with $C = 2$. Use the data file **data.mat** or

`data.dat`. True label of the data are also provided in `label.mat` and `label.dat`.

The matrix `images` is of size 784-by-1990, i.e., there are 1990 images in total, and each column of the matrix corresponds to one image of size 28-by-28 pixels (the image is vectorized; the original image can be recovered by mapping the vector into a matrix).

First, use PCA to reduce the dimensionality of the data before applying to EM. We will put all “6” and “2” digits together, to project the original data into 4-dimensional vectors.

Now implement EM algorithm for the projected data (with 4-dimensions).

Use the full set of data from the provided data files for both training and testing.

1. (10 points) Implement EM algorithm yourself. Use the following initialization
 - initialization for mean: random Gaussian vector with zero mean
 - initialization for covariance: generate two Gaussian random matrix of size n -by- n : S_1 and S_2 , and initialize the covariance matrix for the two components are $\Sigma_1 = S_1 S_1^T + I_n$, and $\Sigma_2 = S_2 S_2^T + I_n$, where I_n is an identity matrix of size n -by- n .

Plot the log-likelihood function versus the number of iterations to show your algorithm is converging.

2. (10 points) Report the fitted GMM model when EM has terminated in your algorithms as follows:
 - The numerical weights for each component.
 - The mean of each component by mapping it back to the original space and reformatting the vectors into 28-by-28 matrices. These should be displayed as images, ideally corresponding to a kind of “average” of the images.
 - Two 4-by-4 covariance matrices by visualizing their intensities (i.e. a gray-scaled image or heatmap).
3. (5 points) Use the τ_k^i to infer the labels of the images, and compare with the true labels. Report the mis-classification rate (1 - Accuracy) for digits “2” and “6” respectively. Perform K -means clustering with $K = 2$ using the same projected data you used for EM. You may call a package or use the code from your previous homework. Find out the mis-classification rate for digits “2” and “6” respectively, and compare with GMM. Which one achieves the better performance?

	“2”	“6”
Kmeans		
GMM		

Explain when comparing K-means and GMM, why do you see one of them outperform the other?

3. Mixture-of-Experts and connection to GMM and EM (25 points)

Mixture-of-Experts (MoE) is a popular stylized model to explain how Large Language Models (LLM) can use different computations for different tokens. This question explores the connection of MoE to Gaussian mixture models (GMM) and the EM algorithm.

In an MoE model, we have K experts and a gate (router). For each input representation $x \in \mathbb{R}^d$, the gate outputs mixture weights $\pi_k(x)$ over experts. Introduce a latent expert index $z \in \{1, \dots, K\}$ indicating which expert generated the output. Consider the following probabilistic MoE regression model for paired data $\{(x_i, y_i)\}_{i=1}^n$:

Gate (softmax).

$$\pi_k(x; \alpha) = \mathbb{P}(z = k | x) = \frac{\exp(a_k^\top x)}{\sum_{j=1}^K \exp(a_j^\top x)}, \quad k = 1, \dots, K.$$

Experts (Gaussian linear regressors, shared variance).

$$p(y | x, z = k; b_k, \sigma^2) = \mathcal{N}(y; b_k^\top x, \sigma^2), \quad k = 1, \dots, K.$$

Let $\theta = (\alpha, \{b_k\}_{k=1}^K, \sigma^2)$.

- (10 points) Write the conditional mixture likelihood $p(y_i | x_i; \theta)$. Then derive the posterior responsibility

$$\gamma_{ik} = \mathbb{P}(z_i = k | x_i, y_i; \theta^{(t)})$$

as an explicit normalized expression involving $\pi_k(x_i; \alpha^{(t)})$ and $\mathcal{N}(y_i; b_k^{(t)\top} x_i, \sigma^{2(t)})$.

- (10 points) Assume α and σ^2 are fixed. Show that the M-step update for each b_k is a weighted least squares problem

$$b_k^{(t+1)} = \arg \min_{b \in \mathbb{R}^d} \sum_{i=1}^n \gamma_{ik} (y_i - b^\top x_i)^2,$$

and write the corresponding normal equations.

- (5 points) Connect back to the GMM in the lecture. Consider the special case where the gate does not depend on x , so $\pi_k(x; \alpha) \equiv \pi_k$, and each expert mean does not depend on x , so $b_k^\top x \equiv \mu_k$. Show that the model reduces to a standard 1D K -component GMM for y ,

$$p(y) = \sum_{k=1}^K \pi_k \mathcal{N}(y; \mu_k, \sigma^2),$$

and explicitly identify the parameter correspondence between (π_k, μ_k, σ^2) and (α, b_k, σ^2) .

4. Optimization (20 points)

Consider a simplified logistic regression problem. Given m training samples (x^i, y^i) , $i = 1, \dots, m$. The data $x^i \in \mathbb{R}$, and $y^i \in \{0, 1\}$. To fit a logistic regression model for classification, we solve the following optimization problem, where $\theta \in \mathbb{R}$ is a parameter we aim to find:

$$\max_{\theta} \ell(\theta), \tag{1}$$

where the log-likelihood function

$$\ell(\theta) = \sum_{i=1}^m \{ -\log(1 + \exp\{-\theta^T x^i\}) + (y^i - 1)\theta^T x^i \}.$$

1. (5 points) Show step-by-step mathematical derivation for the gradient of the cost function $\ell(\theta)$ in (1).
2. (5 points) Write a pseudo-code for performing **gradient descent** to find the optimizer θ^* . This is essentially what the training procedure does.
3. (5 points) Write the pseudo-code for performing the **stochastic gradient descent** algorithm to solve the training of logistic regression problem (1). Please explain the difference between gradient descent and stochastic gradient descent for training logistic regression.
4. (5 points) We will **prove that the training problem in basic logistic regression problem is concave**. Derive the Hessian matrix of $\ell(\theta)$ and based on this, show the training problem (1) is concave. Explain why the problem can be solved efficiently and gradient descent will achieve a unique global optimizer, as we discussed in class.

5. Bayes Classifier for spam filtering (25)

In this problem, we will use the Bayes Classifier algorithm to fit a spam filter by hand. This will enhance your understanding to the Bayes classifier and build intuition. This question does not involve any programming but only derivation and hand calculation. Tools can be used (Python, Excel, Etc.) but all calculations and derivations must still be provided in your report.

Spam filters are used in all email services to classify received emails as “Spam” or “Not Spam”. A simple approach involves maintaining a vocabulary of words that commonly occur in “Spam” emails and classifying an email as “Spam” if the number of words from the dictionary that are present in the email is over a certain threshold. We are given the vocabulary consists of 15 words

$V = \{\text{win, free, nasa, rocket, launch, moon, lunar, artemis, crew, mission, orbit, capsule, spacecraft, astronaut}\}$

We will use V_i to represent the i th word in V . Use the training set provided in the table below.

Spam	Non Spam
win free nasa rocket	crew mission lunar orbit moon
launch moon rocket win nasa	artemis capsule spacecraft astronaut
free lunar launch artemis win	lunar orbit mission free nasa
	artemis crew launch capsule

Table 1: Spam and Non-Spam Messages

Recall that the Naive Bayes classifier assumes the probability of an input depends on its input feature. The feature for each sample is defined as $x^{(i)} = [x_1^{(i)}, x_2^{(i)}, \dots, x_d^{(i)}]^T$, $i = 1, \dots, m$ and the class of the i th sample is $y^{(i)}$. In our case the length of the input vector is $d = 15$, which is equal to the number of words in the vocabulary V . Each entry $x_j^{(i)}$ is equal to the number of times word V_j occurs in the i -th message.

- (5 points) Calculate class prior $\mathbb{P}(y = 0)$ and $\mathbb{P}(y = 1)$ from the training data, where $y = 0$ corresponds to spam messages, and $y = 1$ corresponds to non-spam messages. Note that these class prior essentially corresponds to the frequency of each class in the training sample. Write down the feature vectors for each spam and non-spam messages.
- (10 points) Assuming the keywords follow a multinomial distribution, the likelihood of a sentence with its feature vector x given a class c is given by

$$\mathbb{P}(x|y = c) = \frac{n!}{x_1! \cdots x_d!} \prod_{k=1}^d \theta_{c,k}^{x_k}, \quad c = \{0, 1\}$$

where $n = x_1 + \cdots + x_d$, $0 \leq \theta_{c,k} \leq 1$ is the probability of word k appearing in class c , which satisfies

$$\sum_{k=1}^d \theta_{c,k} = 1, \quad c = \{0, 1\}.$$

Given this, the complete log-likelihood function for our training data is given by

$$\ell(\theta_{0,1}, \dots, \theta_{0,d}, \theta_{1,1}, \dots, \theta_{1,d}) = \sum_{i=1}^m \sum_{k=1}^d x_k^{(i)} \log \theta_{y^{(i)},k}$$

(In this example, $m = 7$.) Derive the maximum likelihood estimates of $\theta_{0,1}$, $\theta_{0,4}$, $\theta_{1,3}$, $\theta_{1,12}$ by maximizing the log-likelihood function above. Note: We are solving a constrained maximization problem. Consider the constraints governing the spam and non-spam word probabilities separately, and introduce the appropriate Lagrangian multipliers / Lagrangian functions for each. You must state your complete objective function before proceeding with optimization to receive full credit.

3. (5 points) Given a test message “free nasa lunar“, using the Naive Bayes classifier that you have trained in Part (a)-(b), to calculate the posterior and decide whether it is spam or not spam. Derivations must be shown in your report.

6. De-bias review system using EM (Bonus, 12 points)

In this question, we will develop an algorithm to remove individual reviewer's bias from their score. Consider the following problem. There are P papers submitted to a machine learning conference. Each of R reviewers reads each paper, and gives it a score indicating how good he/she thought that paper was. We let $x^{(pr)}$ denote the score that reviewer r gave to paper p . A high score means the reviewer liked the paper, and represents a recommendation from that reviewer that it be accepted for the conference. A low score means the reviewer did not like the paper.

We imagine that each paper has some “intrinsic” true value that we denote by μ_p , where a large value means it's a good paper. Each reviewer is trying to estimate, based on reading the paper, what μ_p is; the score reported $x^{(pr)}$ is then reviewer r 's guess of μ_p .

However, some reviewers are just generally inclined to think all papers are good and tend to give all papers high scores; other reviewers may be particularly nasty and tend to give low scores to everything. (Similarly, different reviewers may have different amounts of variance in the way they review papers, making some reviewers more consistent/reliable than others.) We let ν_r denote the “bias” of reviewer r . A reviewer with bias ν_r is one whose scores generally tend to be ν_r higher than they should be.

All sorts of different random factors influence the reviewing process, and hence we will use a model that incorporates several sources of noise. Specifically, we assume that reviewers's scores are generated by a random process given as follows:

$$\begin{aligned}y^{(p)} &\sim \mathcal{N}(\mu_p, \sigma_p^2) \\z^{(r)} &\sim \mathcal{N}(\nu_r, \tau_r^2) \\x^{(pr)}|y^{(p)}, z^{(r)} &\sim \mathcal{N}(y^{(p)} + z^{(r)}, \sigma^2).\end{aligned}$$

The variables $y^{(p)}$ and $z^{(r)}$ are independent; the variables (x, y, z) for different paper-reviewer pairs are also jointly independent. Also, we only ever observe the $x^{(pr)}$ s; thus, the $y^{(p)}$ s and $z^{(r)}$ s are all latent random variables.

We would like to estimate the parameters μ_p , σ_p^2 , ν_r , τ_r^2 . If we obtain good estimates of the papers “intrinsic values” μ_p , these can then be used to make acceptance/rejection decisions for the conference.

We will estimate the parameters by maximizing the marginal likelihood of the data $\{x^{(pr)}; p = 1, \dots, P, r = 1, \dots, R\}$. This problem has latent variables $y^{(p)}$ s and $z^{(r)}$ s, and the maximum likelihood problem cannot be solved in closed form. So, we will use EM.

Your task is to derive the EM update equations. For simplicity, you need to treat only $\{\mu_p, \sigma_p^2; p = 1 \dots, P\}$ and $\{\nu_r, \tau_r^2; r = 1 \dots R\}$ as parameters, i.e. treat σ^2 (the conditional variance of $x^{(pr)}$ given $y^{(p)}$ and $z^{(r)}$) as a fixed, known constant.

1. Derive the E-step (5 points)

- 1.1. The joint distribution $p(y^{(p)}, z^{(r)}, x^{(pr)})$ has the form of a multivariate Gaussian density. Find its associated mean vector and covariance matrix in terms of the

parameters μ_p , σ_p^2 , ν_r , τ_r^2 and σ^2 . [Hint: Recognize that $x^{(pr)}$ can be written as $x^{(pr)} = y^{(p)} + z^{(r)} + \epsilon^{(pr)}$, where $\epsilon^{(pr)} \sim \mathcal{N}(0, \sigma^2)$ is independent Gaussian noise.

- 1.2. Derive an expression for $Q_{pr}(\theta'|\theta) = \mathbb{E}[\log p(y^{(p)}, z^{(r)}, x^{(pr)})|x^{(pr)}, \theta]$ using the conditional distribution $p(y^{(p)}, z^{(r)}|x^{(pr)})$ (E-step) (Hint, you may use the rules for conditioning on subsets of jointly Gaussian random variables.)
2. (5 points) Derive the M-step to update the parameters μ_p , σ_p^2 , ν_r , and τ_r^2 . [Hint: It may help to express an approximation to the likelihood in terms of an expectation with respect to $(y^{(p)}, z^{(r)})$ drawn from a distribution with density $Q_{pr}(y^{(p)}, z^{(r)})$.]
3. (2 points) If a reviewer has very large estimated τ_r^2 , how should their scores be weighted in decision-making?

Remark: John Platt (whose SMO algorithm you've seen) implemented a method quite similar to this one to estimate the papers' true scores. (There, the problem was a bit more complicated because not all reviewers reviewed every paper, but the essential ideas are the same.) Because the model tried to estimate and correct for reviewers' biases, its estimates of the paper's value were significantly more useful for making accept/reject decisions than the reviewers' raw scores for a paper.